

A scientist of many talents, Galileo Galilei made key discoveries that transformed our view of our place in the Universe. He improved the telescope, showed that the Sun was the center of the solar system, and revolutionized theories about the forces of motion, which had been unchanged since Aristotle's time.

Born in Pisa, Italy, Galileo Galilei was the son of a musician and the eldest of six children. He initially wanted to become a monk, but his father wanted him to be a doctor, so he went to study medicine at the University of Pisa. However, while there, he spent most of his time studying mathematics, which he found much more interesting. He went on to become professor of mathematics at Pisa at the age of 25, then later at the University of Padua.

During his time as a student, he watched the big bronze lamp in Pisa Cathedral swinging back and forth. Galileo timed the swings using his pulse and noticed that each swing, however long or short, took the same time. Realizing the pendulum is a regular timekeeper, he invented a simple device for measuring pulses: a small pendulum with a length of string that could be adjusted so that the swing matched the speed of the person's pulse.

Galileo began to conduct experiments to find out why objects move or stop moving. At the time, scientists still believed what Aristotle had said 2,000

MILESTONES

CREATES TELESCOPE

Designs and builds his own telescope in 1609, which reveals the Moon and planets in unprecedented detail.

ASTRONOMICAL FINDS

In 1610, discovers the moons of Jupiter and writes about his observations in *The Starry Messenger*.

TRIED FOR HERESY

Ordered by the Church in 1616 to retract his claim that the Earth is not the center of the Universe.

HOUSE ARREST

Tried for heresy again by the Inquisition in 1633. Spends rest of his life under house arrest.

Galileo was accused of heresy for his belief that the planets orbit the Sun. But in 1992, three and a half centuries later, the Pope officially admitted that Galileo had been right.

